z-Transform

--Definition
--ROC (Region of Converges)
--z-Transform Properties
--Transfer Function

z-Transform

- In continuous signal system, we use S-Transform and FT as the tools to process problems in the transform domain; so in discrete signal system, we use z-Transform and DFT.
- z-Transform can make the solution for discrete time systems very simple.

z-Transform

- The DTFT provides a frequency-domain representation of discrete-time signals and LTI discrete-time systems
- Because of the convergence condition, in many cases, the DTFT of a sequence may not exist.

MOOC

6.1 Definition and Properties key point

- DTFT defined by：

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

leads to the z-transform。

- z-transform may exist for many sequences for which the DTFT does not exist。

6.1 Definition and Properties key point

- For a given sequence g[n], its z-transform G(z) is defined as:

$$G(z) = \sum_{n=-\infty}^{\infty} g[n] z^{-n}$$

where $z = Re(z) + jIm(z)$ is a complex variable.

MOOC

6.1 Definition and Properties

key point

- If we let $z=re^{j\omega}$, then the z-transform reduces to:

$$G(re^{j\omega}) = \sum_{n=-\infty}^{\infty} g[n]r^{-n}e^{-j\omega n}$$

- For r = 1 (i.e., |z| = 1), z-transform reduces to its DTFT, provided the latter exists. The contour |z| = 1 is a circle in the z-plane of unity radius and is called the unit circle.

MOOC

6.1 Definition and Properties

key point

- Like the DTFT, there are conditions on the convergence of the infinite series :

$$\sum_{n=-\infty}^{\infty} g[n]z^{-n}$$

- For a given sequence, the set R of values of z for which its z-transform converges is called the **region of convergence (ROC)**

MOOC

6.1 Definition and Properties

key point

- From our earlier discussion on the uniform convergence of the DTFT, it follows that the series :

$$G(re^{j\omega}) = \sum_{n=-\infty}^{\infty} g[n]r^{-n}e^{-j\omega n}$$

converges if {g[n]r-n} is absolutely summable, i.e., if :

$$\sum_{n=-\infty}^{\infty} \left| g[n]r^{-n} \right| < \infty$$

MOOC

6.1 Definition and Properties

key point

- In general, the ROC of a z-transform of a sequence g[n] is an annular region of the z-plane:

$$R_{g^-} < |z| < R_{g^+}$$

where

$$0 \le R_{g^-} < R_{g^+} \le \infty$$

- Note: The z-transform is a form of a Laurent series and is an analytic function at every point in the ROC。

MOOC

6.1 Definition and Properties key point

- Example - Determine the z-transform X(z) of the causal sequence $x[n]=\alpha^n\mu[n]$ and its ROC.
- Now
$$X(z)=\sum_{n=-\infty}^{\infty}\alpha^n\mu[n]z^{-n}=\sum_{n=0}^{\infty}\alpha^n z^{-n}$$

The above power series converges to:

$$X(z)=\frac{1}{1-\alpha z^{-1}},\quad \text{for } \left|\alpha z^{-1}\right|<1$$

ROC is the annular region |z| > |a|.

MOOC

6.1 Definition and Properties

key point

- Example - The z-transform μ(z) of the unit step sequence μ[n] can be obtained from:

$$X(z) = \frac{1}{1-\alpha z^{-1}}, \quad \text{for } \left|\alpha z^{-1}\right| < 1$$

by setting a = 1,

$$\mu(z) = \frac{1}{1-z^{-1}}, \quad \text{for } \left|z^{-1}\right| < 1$$

ROC is the annular region $1 < |z| < \infty$.

MOOC

6.1 Definition and Properties

key point

- Example - Consider the anti-causal sequence:

$$y[n] = -\alpha^n \mu[-n-1]$$

  - Its z-transform is given by:

$$Y(z) = \sum_{n=-\infty}^{-1} -\alpha^n z^{-n} = -\sum_{m=1}^{\infty} \alpha^{-m} z^m$$

$$= \frac{1}{1-\alpha z^{-1}}, \ for |z| < |\alpha|$$

ROC is the annular region $|z| < |\alpha|$

MOOC

6.1 Definition and Properties

key point

- Note: The unit step sequence $\mu[n]$ is not absolutely summable, and hence its DTFT does not converge uniformly.
- Note: Only way an unique sequence can be associated with a z-transform is by specifying its ROC.

Commonly Used z-transform

Sequence	z-Transform	ROC
$\delta[n]$	1	All values of z
$\mu[n]$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$\alpha^n\mu[n]$	$\dfrac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$
$(r^n\cos\omega_o n)\mu[n]$	$\dfrac{1-(r\cos\omega_o)z^{-1}}{1-(2r\cos\omega_o)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$(r^n\sin\omega_o n)\mu[n]$	$\dfrac{(r\sin\omega_o)z^{-1}}{1-(2r\cos\omega_o)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$

MOOC

6.2 Rational z-Transform

- In the case of LTI discrete-time systems we are concerned with in this course, all pertinent z-transforms are rational functions of z^{-1}.
- That is, they are ratios of two polynomials in z^{-1} :

$$G(z) = \frac{P(z)}{D(z)} = \frac{p_0 + p_1 z^{-1} + \cdots + p_{M-1} z^{-(M-1)} + p_M z^{-M}}{d_0 + d_1 z^{-1} + \cdots + d_{N-1} z^{-(N-1)} + d_N z^{-N}}$$

MOOC

6.2 Rational z-Transform

- A rational z-transform can be alternately written in factored form as:

$$G(z)=\frac{p_0\prod_{\ell=1}^{M}(1-\xi_\ell z^{-1})}{d_0\prod_{\ell=1}^{N}(1-\lambda_\ell z^{-1})}$$

$$=z^{(N-M)}\frac{p_0\prod_{\ell=1}^{M}(z-\xi_\ell)}{d_0\prod_{\ell=1}^{N}(z-\lambda_\ell)}$$

MOOC

6.2 Rational z-Transform

- At a root $z=\xi_l$ of the numerator polynomial $G(\xi_l)=0$ and as a result, these values of z are known as the **zeros** of G(z).
- At a root $z=\lambda_l$ of the denominator polynomial $G(\lambda_l)\rightarrow\infty$, and as a result, these values of z are known as the **poles** of G(z).

MOOC

6.2 Rational z-Transform

- Consider:

$$G(z) = z^{(N-M)} \frac{p_0 \prod_{\ell=1}^{M} (z - \xi_\ell)}{d_0 \prod_{\ell=1}^{N} (z - \lambda_\ell)}$$

Note G(z) has M finite zeros and N finite poles:

- If N > M there are additional N - M zeros at z = 0 (the origin in the z-plane)
- If N < M there are additional M - N poles at z = 0.

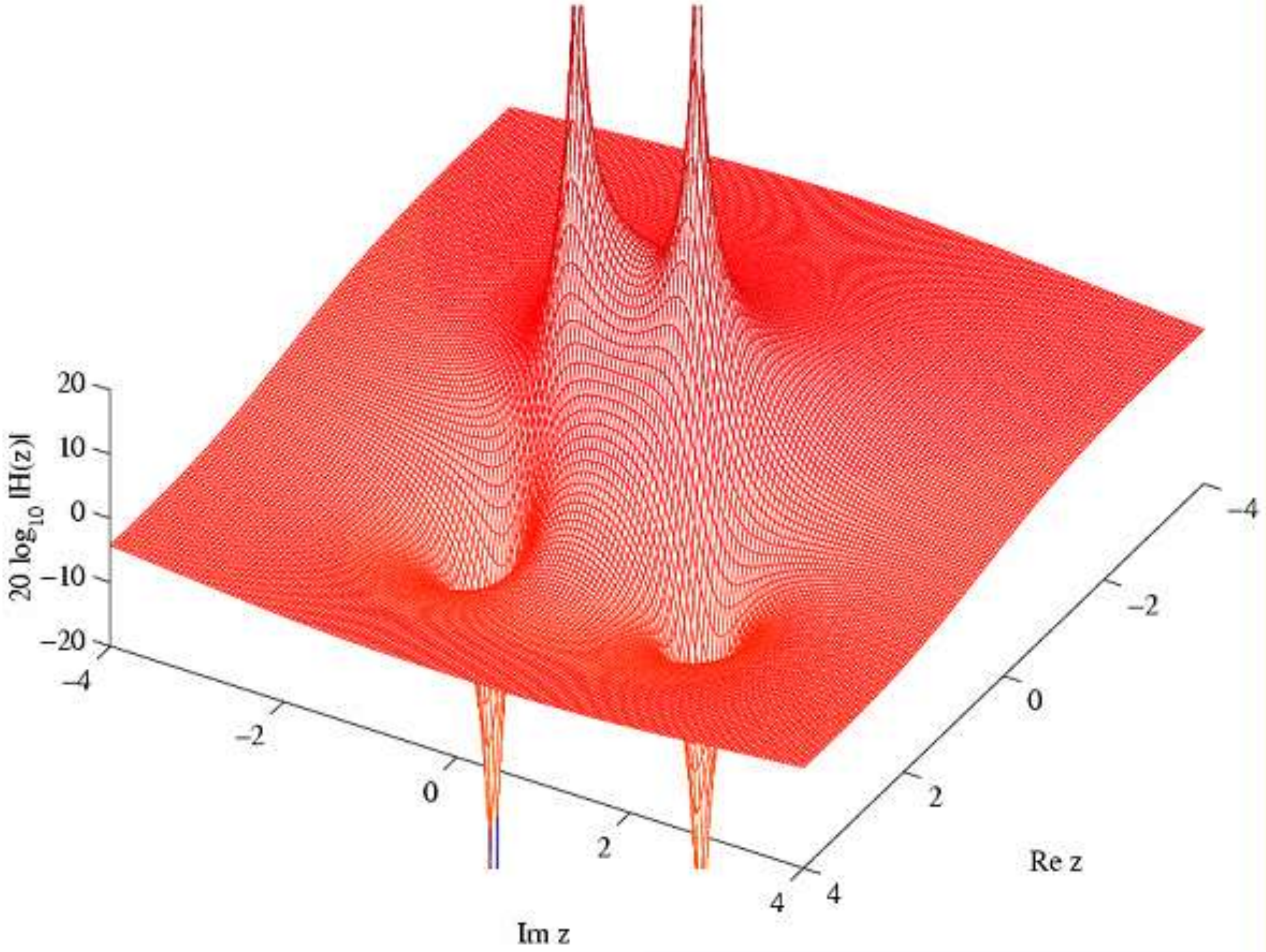
MOOC

6.2 Rational z-Transform

- A physical interpretation of the concepts of poles and zeros can be given by plotting the log-magnitude $20\log_{10}|G(z)|$ as shown on next figure for:

$$G(z)=\frac{1-2.4z^{-1}+2.88z^{-2}}{1-0.8z^{-1}+0.64z^{-2}}$$

6.2 Rational z-Transform



poles $z=0.4\pm j0.6928$,
zeros $z=1.2\pm j1.2$.

6.3 ROC of a Rational z-transform

- ROC of a z-transform is an important concept.
- Without the discussion of the ROC, there is no unique relationship between a sequence and its z-transform.
- Hence, the z-transform must always be specified with its ROC.

6.3 ROC of a Rational z-transform

- Moreover, if the ROC of a z-transform includes the unit circle, the DTFT of the sequence is obtained by simply evaluating the z-transform on the unit circle.
- ROC of z-transform of the impulse sequence of a causal, stable LTI discrete time system.

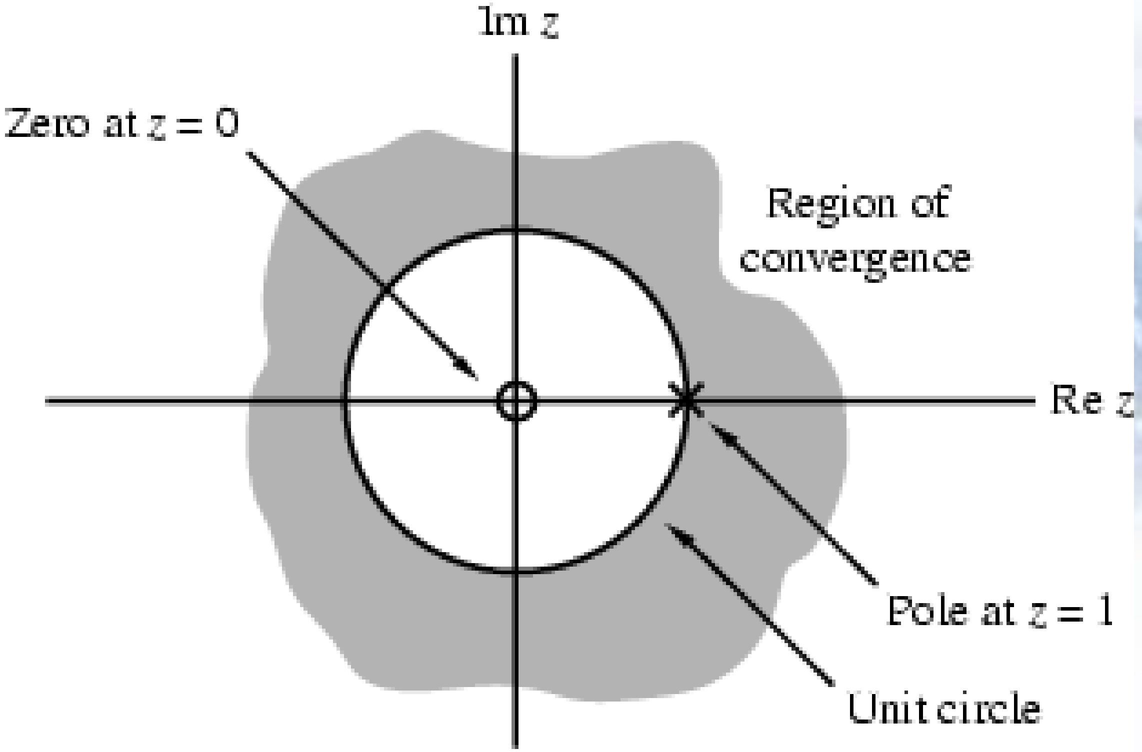
6.3 ROC of a Rational z-transform

- The ROC of a rational z-transform is bounded by the locations of its poles.
- To understand the relationship between the poles and the ROC, it is instructive to examine the pole-zero plot of a z-transform.

MOOC

6.3 ROC of a Rational z-transform

- Consider again the pole-zero plot of the z-transform $\mu(z)$.



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6.3 ROC of a Rational z-transform

- A sequence can be one of the following types: finite-length, right-sided, left-sided and two-sided.
- In general, the ROC depends on the type of the sequence of interest.

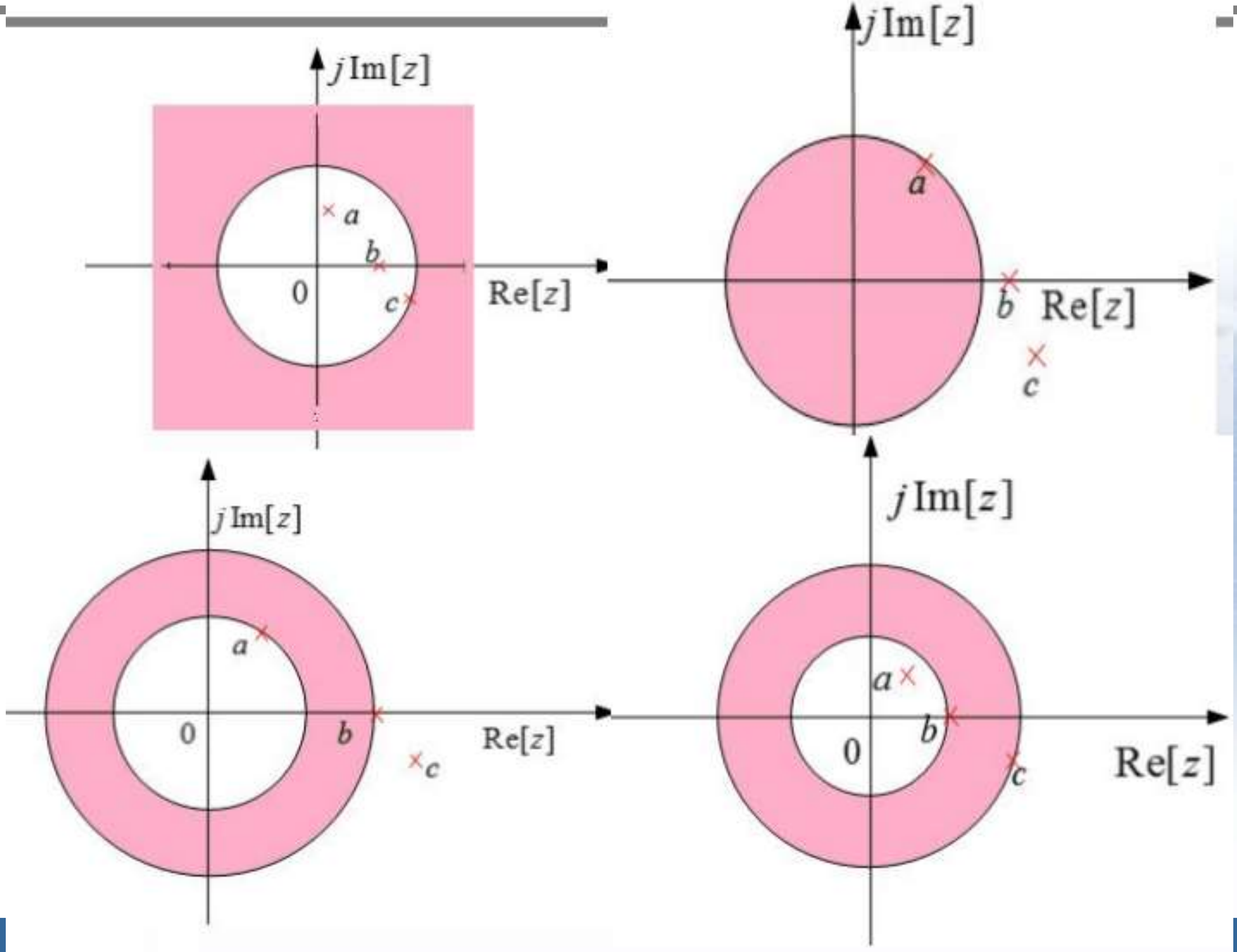
MOOC

6.3 ROC of a Rational z-transform

- The ROC of a rational z-transform is bounded by the locations of its poles.
- To understand the relationship between the poles and the ROC, it is instructive to examine the pole-zero plot of a z-transform.

MOOC

6.3 ROC of a Rational z-transform



Three poles and possible ROCs

MOOC

6.3 ROC of a Rational z-transform

- The ROC of a rational z-transform can be easily determined using MATLAB

  [z, p, k] = **tf2zp**(num, den)

  determines the zeros, poles, and the gain constant of a rational z-transform with the numerator and denominator coefficients specified by the vector num and den, respectively.

6.3 ROC of a Rational z-transform

[num,den] = zp2tf(z, p, k)

implements the reverse process.

- The factored form of the z-transform can be obtained using

  sos = zp2sos(z, p, k)

- The above statement computes the coefficients of each second-order factor given as an L×6 matrix sos.

6.3 ROC of a Rational z-transform

$$sos = \begin{bmatrix} b_{01} & b_{11} & b_{21} & a_{01} & a_{11} & a_{12} \\ b_{02} & b_{12} & b_{22} & a_{02} & a_{12} & a_{22} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ b_{0L} & b_{1L} & b_{2L} & a_{0L} & a_{1L} & a_{2L} \end{bmatrix}$$

$$G(z) = \prod_{k=1}^{L} \frac{b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2}}{a_{0k} + a_{1k} z^{-1} + a_{2k} z^{-2}}$$

6.3 ROC of a Rational z-transform

- The pole-zero plot is determined using the function zplane.
- The z-transform can be either described in terms of its zeros and poles: zplane(zeros,poles).
- or, it can be described in terms of its numerator and denominator coefficients: zplane(num,den) .

Please look at P287-288 examples about z-transform using MATLAB .

6.4 The Inverse z-Transform key point

- General Expression
- Inverse z-Transform by Table Look-up Method
- Inverse z-Transform by Partial-Fraction Expansion
- Inverse z-Transform via Long Division
- MATALB Computation

6.4.1 General Expression

key point

- Inverse ZT is given as:

$$g[n] = \frac{1}{2\pi j}\oint_{C'} G(z) z^{n-1} dz$$

where C' is a counterclockwise contour of integration defined by $|z|=r$.

6.4.1 General Expression

key point

- But the contour integral is difficult to calculate.
- The contour integral can be evaluated using the Cauchy's residue theorem resulting in:

$$g[n]=\left[\begin{array}{c}\text{residues of } G(z)z^{n-1} \\ \text{at the poles inside } C\end{array}\right]$$

General Expression

key point

- Based on the residue theorem, if

$$F(z)=G(z)z^{n-1}$$

is continuous in the contour C, and there are K poles z_k inside C, M poles z_m outside C.

$$\frac{1}{2\pi j}\oint_C G(z)z^{n-1}dz=\sum_k \mathrm{Re}\,s[G(z)z^{n-1}]_{z=z_k}$$

C is counterclockwise.

$$\frac{1}{2\pi j}\oint_C G(z)z^{n-1}dz=\sum_m \mathrm{Re}\,s[G(z)z^{n-1}]_{z=z_m}$$

C is clockwise.

6.4.1 General Expression

key point

- Inverse z-Transform Expression:

$$g[n]=\frac{1}{2\pi j}\oint_C G(z)z^{n-1}dz=\sum_k \operatorname{Res}[G(z)z^{n-1}]_{z=z_k}$$

$$g[n]=\frac{1}{2\pi j}\oint_C G(z)z^{n-1}dz=-\sum_m \operatorname{Res}[G(z)z^{n-1}]_{z=z_m}$$

Where C is counterclockwise.

How To Solve the Residue at Any Pole z_r?

- If z_r is a **single pole**, then:

$$\mathrm{Res}[G(z)z^{n-1}]_{z=z_r} = [(z-z_r)G(z)z^{n-1}]_{z=z_r}$$

If z_r is a **multiple pole(M order)**, then:

$$\mathrm{Res}[G(z)z^{n-1}]_{z=z_r} = \frac{1}{(M-1)!}\frac{d^{M-1}}{dz^{M-1}}[(z-z_r)^M G(z)z^{n-1}]_{z=z_r}$$

Other Caculation Methods key point

- Table Look-up Method
- Partial Fraction Expansion
- Inverse z-Transform via Long Division

MOOC

6.4.2 Table Look-up Method

- Illustrate this method in Example 6.12:

$$H(z)=\frac{0.5z}{z^2-z+0.25},|z|>0.5$$

$$H(z)=\frac{0.5z}{(z-0.5)^2}=\frac{0.5z^{-1}}{(1-0.5z^{-1})^2}$$

From table 6.1, we get:

$$n\alpha^n\mu[n]\xleftrightarrow{z}\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2},|z|>|\alpha|$$

So:

$$h[n]=n(0.5)^n\mu[n]$$

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6.4.3 Partial-Fraction Expansion

- In practical application, the X(z) is the rational fraction of z.

$$X(z) = B(z)/A(z) = X_1(z) + X_2(z) + \ldots + X_k(z)$$

Where A(z),B(z) are polynomials of z.
So:

$$x[n] = Z^{-1}[X(z)] = Z^{-1}[X_1(z)] + Z^{-1}[X_2(z)] + \ldots + Z^{-1}[X_k(z)]$$

Note:you must make each $X_k(z)$ easy to solve its IZT, and pay attention to it's ROC.

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6.4.3 Partial-Fraction Expansion

If: $$X(z)=\frac{B(z)}{A(z)}=\frac{p_0+p_1z^{-1}+\cdots+p_{M-1}z^{-(M-1)}+p_Mz^{-M}}{d_0+d_1z^{-1}+\cdots+d_{N-1}z^{-(N-1)}+d_Nz^{-N}}=\sum_{l=0}^{M-N}\eta_l z^{-l}+\frac{B_1(z)}{A(z)}$$

$$=\sum_{l=0}^{M-N}\eta_l z^{-l}+\frac{B_1(z)}{(1-\lambda_1z^{-1})(1-\lambda_2z^{-1})\cdots(1-\lambda_{N-L}z^{-1})(1-vz^{-1})^L}$$

Then X(z) is expanded as partial-fraction:

$$X(z)=\sum_{l=0}^{M-N}\eta_l z^{-l}+\sum_{l=1}^{N-L}\frac{\rho_l}{1-\lambda_l z^{-1}}+\sum_{i=1}^{L}\frac{\gamma_i}{(1-vz^{-1})^i}$$

Where v is a L-order poles of X(z) ,
each λ_l is a single pole of
X(z)(k=1,2,…,N-L).

MOOC

6.4.3 Partial-Fraction Expansion

- η_l: when M≥N, it exists; M<N, it equals to 0.
- ρ_l: based on residue theorem, we get

$$\rho_l = (1-\lambda_l z^{-1})X(z)|_{z=\lambda_l} = (z-\lambda_l)\frac{X(z)}{z} = \mathrm{Res}[\frac{X(z)}{z}]_{z=\lambda_l}$$

l=1,2,…N-L.

- γ_i: $$\gamma_i = \frac{1}{(-v)^{L-i}}\frac{1}{(L-i)!}\{\frac{d^{L-i}}{d(z^{-1})^{L-i}}[(1-vz^{-1})^L X(z)]\}_{z=v}, i=1,2,\ldots,L$$

MOOC

6.4.3 Partial-Fraction Expansion

- Example - Consider

$$G(z)=\frac{2+0.8z^{-1}+0.5z^{-2}+0.3z^{-3}}{1+0.8z^{-1}+0.2z^{-2}}$$

- By long division we arrive at:

$$G(z)=-3.5+1.5z^{-1}+\frac{5.5+2.1z^{-1}}{1+0.8z^{-1}+0.2z^{-2}}=-3.5+1.5z^{-1}+G_1(z)$$

MOOC

6.4.3 Partial-Fraction Expansion

- Example - Let the z-transform H(z) of a causal sequence h[n] be given by:

$$H(z) = \frac{z(z+2)}{(z-0.2)(z+0.6)} = \frac{1+2z^{-1}}{(1-0.2z^{-1})(1+0.6z^{-1})}$$

- A partial-fraction expansion of H(z) is then of the form:

$$H(z) = \frac{\rho_1}{1-0.2z^{-1}} + \frac{\rho_2}{1+0.6z^{-1}}$$

MOOC

6.4.3 Partial-Fraction Expansion

- **Now:**

$$\rho_1 = (1-0.2z^{-1})H(z)\big|_{z=0.2} = \left.\frac{1+2z^{-1}}{1+0.6z^{-1}}\right|_{z=0.2} = 2.75$$

and:

$$\rho_2 = (1+0.6z^{-1})H(z)\big|_{z=-0.6} = \left.\frac{1+2z^{-1}}{1-0.2z^{-1}}\right|_{z=-0.6} = -1.75$$

MOOC

6.4.3 Partial-Fraction Expansion

- Hence

$$H(z) = \frac{2.75}{1-0.2z^{-1}} - \frac{1.75}{1+0.6z^{-1}}$$

The inverse transform of the above is therefore given by:

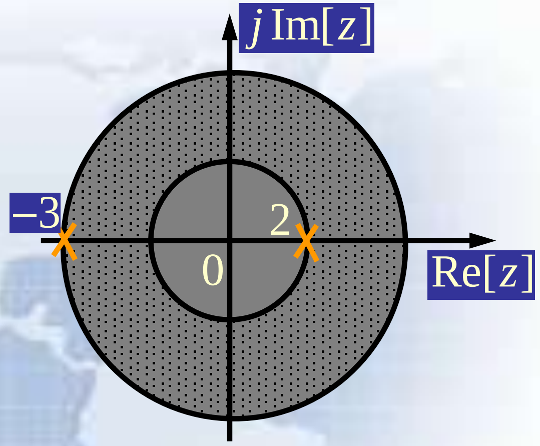
$$h[n] = 2.75(0.2)^n \mu[n] - 1.75(-0.6)^n \mu[n]$$

MOOC

$$\text{E.g: } X(z)=\frac{5z^{-1}}{1+z^{-1}-6z^{-2}},\ 2<|z|<3,\ \text{Find out IZT}$$

$$\text{Solution: } X(z)=\frac{5z^{-1}}{1+z^{-1}-6z^{-2}}=\frac{5z}{z^2+z-6}=\frac{5z}{(z-2)(z+3)}$$

$$\frac{X(z)}{z}=\frac{5}{(z-2)(z+3)}=\frac{A_1}{z-2}+\frac{A_2}{z+3}$$



$$A_1=Res\left[\frac{X(z)}{z}\right]_{z=2}=(z-2)\frac{5}{(z-2)(z+3)}\bigg|_{z=2}=1$$

$$A_2=Res\left[\frac{X(z)}{z}\right]_{z=-3}=(z+3)\frac{5}{(z-2)(z+3)}\bigg|_{z=-3}=-1$$

MOOC

$$\frac{X(z)}{z}=\frac{1}{z-2}+\frac{-1}{z+3}$$

$$X(z)=\frac{z}{z-2}+\frac{-z}{z+3}=\frac{1}{1-2z^{-1}}+\frac{-1}{1+3z^{-1}}$$

$\because 2<|z|<3$

$$ZT[a^n u(n)]=\frac{1}{1-az^{-1}} \quad |z|>|a|$$

$$ZT[a^n u(-n-1)]=\frac{-1}{1-az^{-1}} \quad |z|<|a|$$

$$\frac{1}{1-2z^{-1}} \xrightarrow{|z|>2} 2^n u(n)$$

$$\frac{-1}{1+3z^{-1}} \xrightarrow{|z|<3} (-3)^n u(-n-1)$$

$$\therefore x(n)=2^n u(n)+(-3)^n u(-n-1)$$

MOOC

6.4.4 Partial-Fraction Expansion MATALB Computation

- [r,p,k]= residuez(num,den) develops the partial-fraction expansion of a rational z-transform with numerator and denominator coefficients given by vectors num and den.
- Vector r contains the residues.
- Vector p contains the poles.
- Vector k contains the constants.

6.4.4 Partial-Fraction Expansion MATALB Computation

- [num,den]=residuez(r,p,k) converts a z-transform expressed in a partial-fraction expansion form to its rational form.
- P293-294 Examples.

6.4.5 Inverse z-Transform via Long Division

Because in common condition, $X(z)$ is a rational fraction, we can divide numerator polynomial by denominator polynomial to get the expansion of the power series.

Note: When we use long division methods, we must first estimate the ROC of $x[n]$, then expand $X(z)$ into appropriate $x[n]$.

$$\text{E.g}: X(z)=\frac{z^2}{(4-z)(z-1/4)},\ 1/4<|z|<4,\ \text{It's IZT?}$$

Solution : Roc of X(z) is annular , so x[n] is a two-sided sequence

pole z=1/4 relates to right-sided sequence,
pole z=4 relates to left-sided sequence

Expand X(z) into Partial-Fraction Expansion

$$\frac{X(z)}{z}=\frac{z}{(4-z)(z-1/4)}=\frac{\frac{16}{15}}{4-z}+\frac{\frac{1}{15}}{z-1/4}$$

$$X(z)=\frac{1}{15}\left(\frac{16z}{4-z}+\frac{z}{z-1/4}\right)$$

$$
\begin{array}{r}
4z+z^2+\frac{1}{4}z^3+\cdots \\
4-z\,\big)\overline{16z\qquad\qquad\qquad} \\
\underline{16z-4z^2} \\
4z^2 \\
\underline{4z^2-z^3} \\
z^3 \\
\vdots
\end{array}
$$

$$
\begin{array}{r}
1+\frac{1}{4}z^{-1}+\frac{1}{16}z^{-2}+\cdots \\
z-\frac{1}{4}\,\big)\overline{z\qquad\qquad\qquad} \\
\underline{z-\frac{1}{4}} \\
\frac{1}{4} \\
\underline{\frac{1}{4}-\frac{1}{16}z^{-1}} \\
\frac{1}{16}z^{-1} \\
\vdots
\end{array}
$$

$$X(z)=\frac{1}{15}\left(\cdots+\frac{1}{16}z^{-2}+\frac{1}{4}z^{-1}+1+4z+z^{2}+\frac{1}{4}z^{3}+\cdots\right)$$

$$=\frac{1}{15}\left[\sum_{n=0}^{\infty}\left(\frac{1}{4}\right)^{n}z^{-n}+\sum_{n=-1}^{-\infty}4^{n+2}z^{-n}\right]$$

$$\therefore x(n)=\frac{4^{-n}}{15}u(n)+\frac{4^{n+2}}{15}u(-n-1)$$

6.4.6 Inverse z-Transform Using MATLAB

- The function **impz** can be used to find the inverse of a rational z-transform $G(z)$.
- The function computes the coefficients of the power series expansion of $G(z)$.
- The number of coefficients can either be user specified or determined automatically.

6.5 z-Transform Properties key point

Property	Sequence	z -Transform	ROC
	$g[n]$ $h[n]$	$G(z)$ $H(z)$	$\mathcal{R}_g$ $\mathcal{R}_h$
Conjugation	$g^*[n]$	$G^*(z^*)$	$\mathcal{R}_g$
Time-reversal	$g[-n]$	$G(1/z)$	$1/\mathcal{R}_g$
Linearity	$\alpha g[n] + \beta h[n]$	$\alpha G(z) + \beta H(z)$	Includes $\mathcal{R}_g \cap \mathcal{R}_h$
Time-shifting	$g[n - n_o]$	$z^{-n_o} G(z)$	$\mathcal{R}_g$, except possibly the point $z = 0$ or ∞
Multiplication by an exponential sequence	$\alpha^n g[n]$	$G(z/\alpha)$	$\lvert\alpha\rvert\mathcal{R}_g$
Differentiation of $G(z)$	$ng[n]$	$-z\dfrac{dG(z)}{dz}$	$\mathcal{R}_g$, except possibly the point $z = 0$ or ∞
Convolution	$g[n] \circledast h[n]$	$G(z)H(z)$	Includes $\mathcal{R}_g \cap \mathcal{R}_h$
Modulation	$g[n]h[n]$	$\frac{1}{2\pi j} \oint_C G(v)H(z/v)v^{-1}\,dv$	Includes $\mathcal{R}_g\mathcal{R}_h$

Parseval's relation $\quad \displaystyle\sum_{n=-\infty}^{\infty} g[n]h^*[n] = \frac{1}{2\pi j} \oint_C G(v)H^*(1/v^*)v^{-1}\,dv$

Note: If $\mathcal{R}_g$ denotes the region $R_{g^-} < |z| < R_{g^+}$ and $\mathcal{R}_h$ denotes the region $R_{h^-} < |z| < R_{h^+}$, then $1/\mathcal{R}_g$ denotes the region $1/R_{g^+} < |z| < 1/R_{g^-}$ and $\mathcal{R}_g\mathcal{R}_h$ denotes the region $R_{g^-}R_{h^-} < |z| < R_{g^+}R_{h^+}$.

MOOC

6.5 z-Transform Properties

1.Initial value theorem

For a causal sequence x[n], exist:

$$\lim_{z->\infty} X(z) = x[0]$$

2.Terminal value theorem

For a causal sequence x[n], and the pole of X(z)=Z(x[n]) inside the unit cycle (in unit cycle there are at best single poles), then:

$$\lim_{n->\infty} x[n] = \lim_{z->1}[(z-1)X(z)]$$

6.6 Computation of the Convolution Sum of Finite-Length Sequences

- Tabular methods for the computation of the linear and circular convolution have been outlined.
- Now, we describe alternate methods for the computation of the linear and circular convolution that based on the multiplication of two polynomial.

6.6.1 Linear Convolution

- Let x[n] and h[n] be two sequences of lengths L+1 and M+1, respectively. Their z-transforms, X(z) and H(z) be:

$$X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots + x[L]z^{-L},$$

$$H(z) = h[0] + h[1]z^{-1} + h[2]z^{-2} + \cdots + h[M]z^{-M}.$$

So, the z-transform of linear convolution sequence y[n] is Y(z):

$$Y(z) = X(z)H(z) = y[0] + y[1]z^{-1} + y[2]z^{-2} + \cdots + y[L+M]z^{-(L+M)},$$

And its coefficient is y[n].

6.6.2 Circular Convolution

- Let x[n] and h[n] both be two sequences of degree N-1, Their z-transforms, X(z) and H(z) be:

$$X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots + x[N-1]z^{-(N-1)},$$

$$H(z) = h[0] + h[1]z^{-1} + h[2]z^{-2} + \cdots + h[N-1]z^{-(N-1)}.$$

So, the z-transform of linear convolution sequence y[n] is Y(z):

$$Y_L(z) = y_L[0] + y_L[1]z^{-1} + y_L[2]z^{-2} + \cdots + y_L[2N-2]z^{-(2N-2)}$$

6.6.2 Circular Convolution

- Let $Y_c(z)$ denote the polynomial of degree N-1 whose coefficients is $y_c[n]$, it can be shown that (Problem 6.17) :

$$Y_C(z) = \langle Y_L(z) \rangle_{(z^{-N}-1)}$$

The modulo operation with respect to $z^{-N}=1$ in above equation of $Y_L(z)$.

Please look at P331 about the process example.

6.7 The Transfer Function

- The z-transform of the impulse response of an LTI system, called the **transfer function**, is a polynomial in z^{-1}.
- In most practical cases, the LTI digital filter of interest is characterized by a linear different equation with constant and real coefficient, so it's transfer function is rational z-transform.

6.7.1 Definition

- Origin:

In time domain, use unit sample response h[n] to represent a LTI system:

$$y[n] = x[n] * h[n]$$

To do ZT for both sides, we get:

$$Y(z) = H(z)X(z)$$

$$H(z) = Y(z)/X(z)$$

It's the **transfer function** of a LTI system.

MOOC

6.7.2 Transfer Function Expression

- FIR Digital Filter

In the case of an LTI FIR digital filter, with it’s impulse response h[n] defined for $N_1 \le n \le N_2$, and thus, h[n]=0 for $n < N_1$ and $n > N_2$. Therefore, the transfer function is given by:

$$H(z) = \sum_{n=N_1}^{N_2} h[n] z^{-n}$$

6.7.2 Transfer Function Expression

- Finite-Dimension Linear Time-Invariant IIR Discrete-Time System

Consider an LTI discrete-time system characterized by a difference equation

$$\sum_{k=0}^{N} d_k y[n-k] = \sum_{k=0}^{M} p_k x[n-k]$$

Its transfer function is obtained by taking the z-transform of both sides of the above equation, Thus:

$$H(z) = \frac{\sum_{k=0}^{M} p_k z^{-k}}{\sum_{k=0}^{N} d_k z^{-k}}$$

6.7.2 Transfer Function Expression

$$H(z)=\frac{\sum_{k=0}^{M} p_k z^{-k}}{\sum_{k=0}^{N} d_k z^{-k}} = \frac{p_0}{d_0} z^{(N-M)} \frac{\prod_{m=1}^{M}(z-\xi_m)}{\prod_{k=1}^{N}(z-\lambda_k)}$$

1. $\xi_1,\xi_2,\xi_3,\ldots\ldots,\xi_M$ are the finite zeros, and $\lambda_1,\lambda_2,\ldots\ldots,\lambda_N$ are the finite poles of H(z).

2. If N>M, there are additional (N-M) zeros at z=0 ; if N<M, there are additional (M-N) poles at z=0.

3. For a causal IIR filter, the ROC of the transfer function H(z) is: $|z| > \max_k |\lambda_k|$

6.7.3 Frequency Response from Transfer Function

The relationship between H(z) and H(e^jω).

$$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}} \, .$$

$$H(z) = \frac{p_0}{d_0} \frac{\prod_{m=1}^{M}(1-\xi_m z^{-1})}{\prod_{k=1}^{N}(1-\lambda_k z^{-1})} = \frac{p_0}{d_0} z^{(N-M)} \frac{\prod_{m=1}^{M}(z-\xi_m)}{\prod_{k=1}^{N}(z-\lambda_k)}$$

$$z = e^{j\omega}$$

$$H(e^{j\omega}) = \frac{p_0}{d_0} e^{j\omega(N-M)} \frac{\prod_{k=1}^{M}(e^{j\omega}-\xi_k)}{\prod_{k=1}^{N}(e^{j\omega}-\lambda_k)} = \left|H(e^{j\omega})\right| e^{j\arg[H(e^{j\omega})]}$$

6.7.4 Geometric Interpretation of FR Computation

Use zero-vectors and pole-vectors in z-plane to interpret the frequency response.

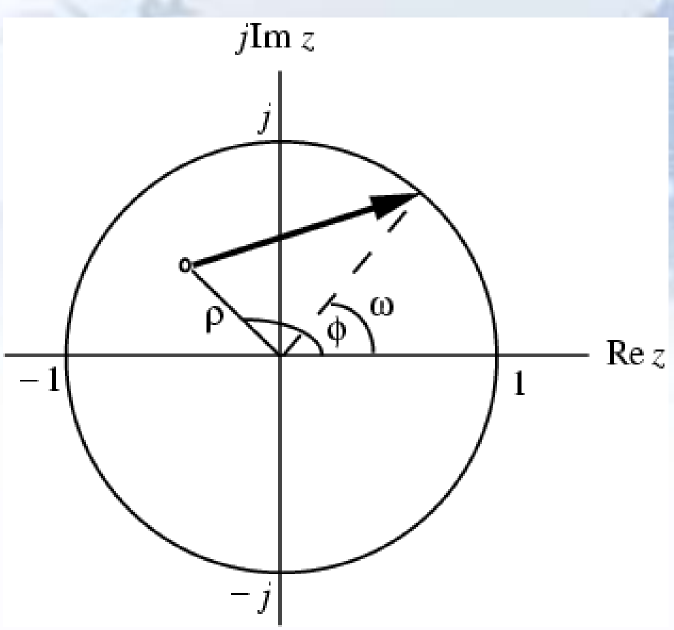
$$\left|H(e^{j\omega})\right| = \left|\frac{p_0}{d_0}\right| \frac{\prod_{k=1}^{M}\left|e^{j\omega} - \xi_k\right|}{\prod_{k=1}^{N}\left|e^{j\omega} - \lambda_k\right|}$$

$$\arg H(e^{j\omega})$$

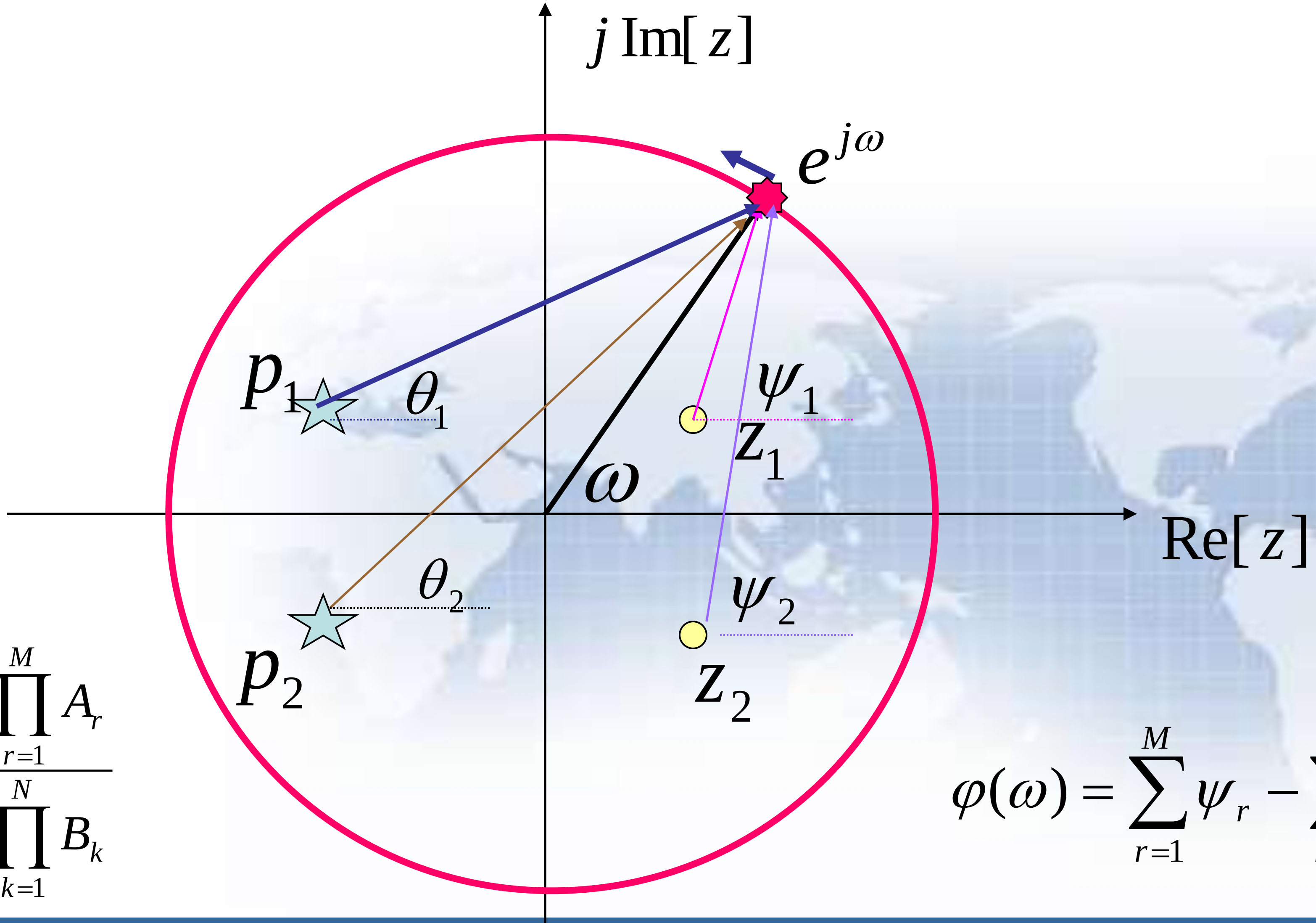
$$= \arg(p_0/d_0) + \omega(N - M) + \sum_{k=1}^{M}\arg(e^{j\omega} - \xi_k) - \sum_{k=1}^{N}\arg(e^{j\omega} - \lambda_k)$$

6.7.4 Geometric Interpretation of FR Computation

- As shown below in the z-plane the factor $(e^{j\omega} - \rho e^{j\phi})$ represents a vector starting at the point $z = \rho e^{j\phi}$ and ending on the unit circle at $z = e^{j\omega}$

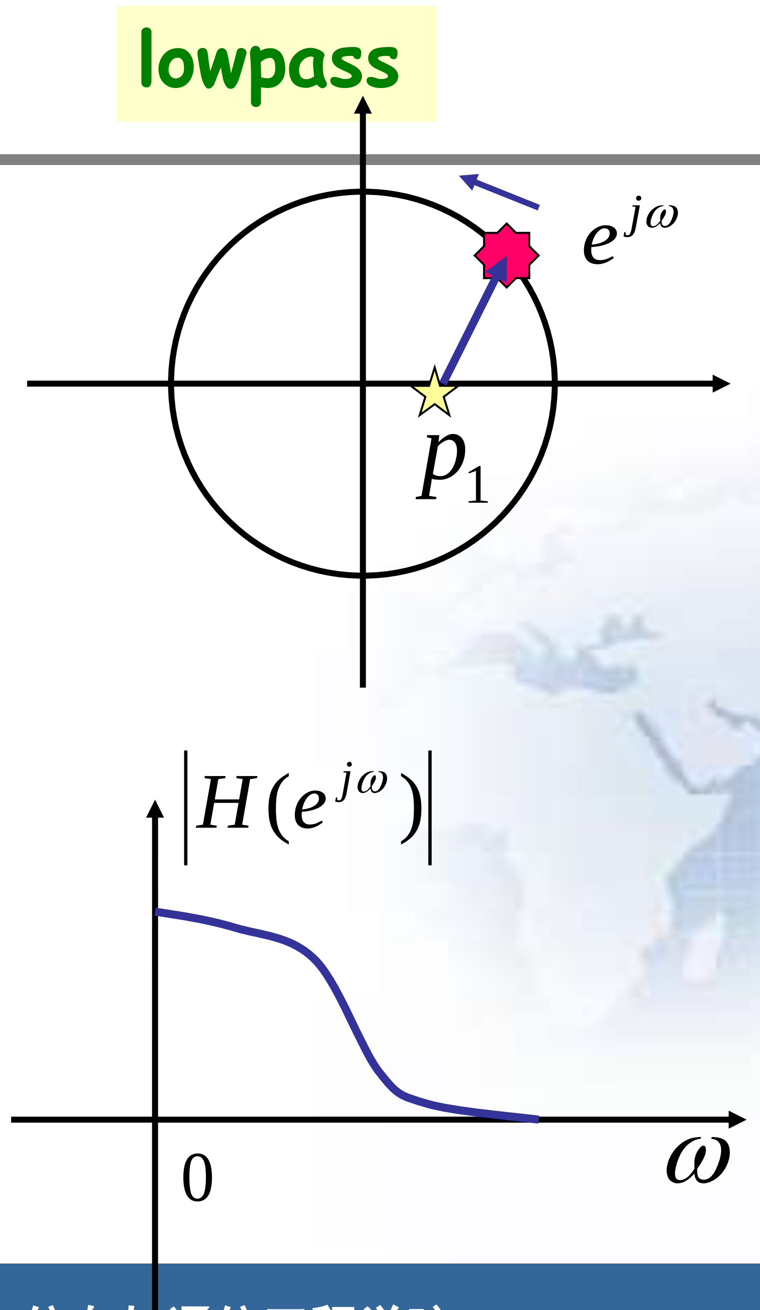


$$H(z)=\left|H(e^{j\omega})\right|e^{j\varphi(\omega)}$$

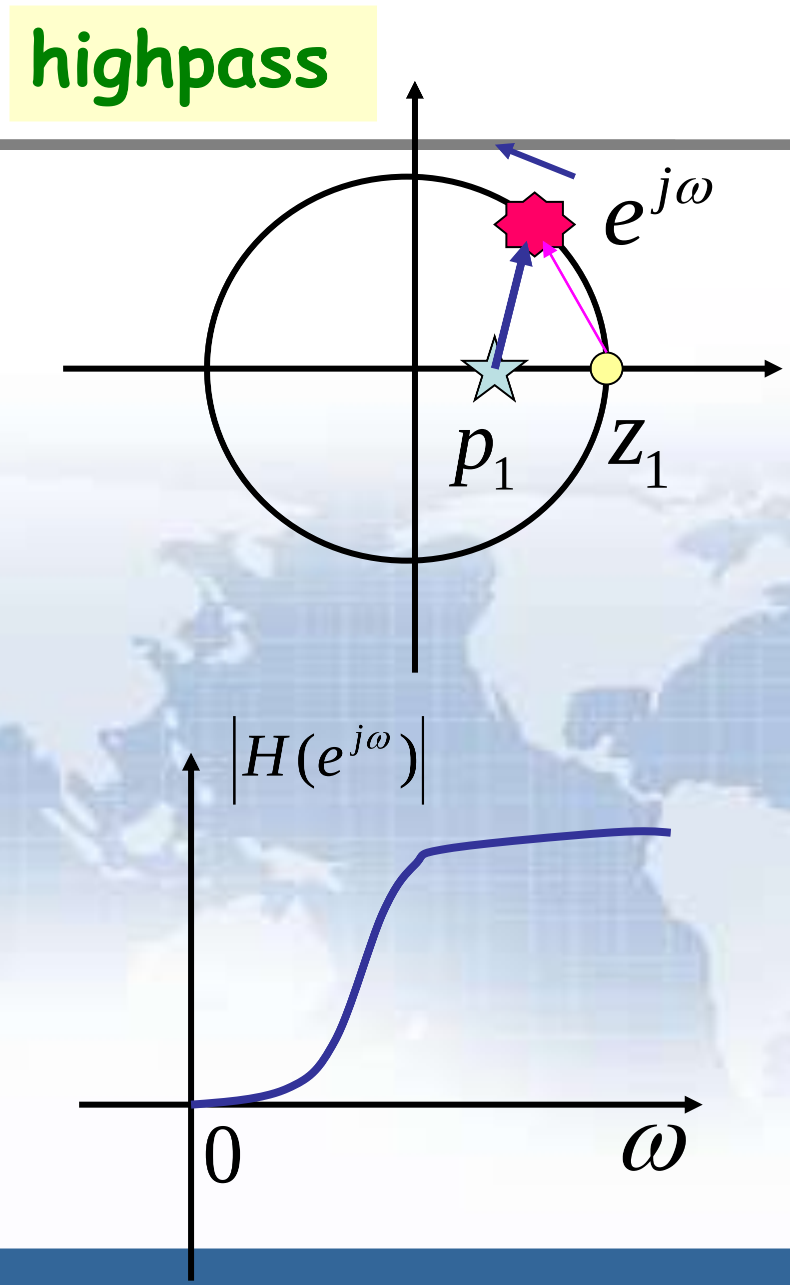


$$\left|H(e^{j\omega})\right|=\frac{\prod_{r=1}^{M}A_r}{\prod_{k=1}^{N}B_k}$$

$$\varphi(\omega)=\sum_{r=1}^{M}\psi_r-\sum_{k=1}^{N}\theta_k$$



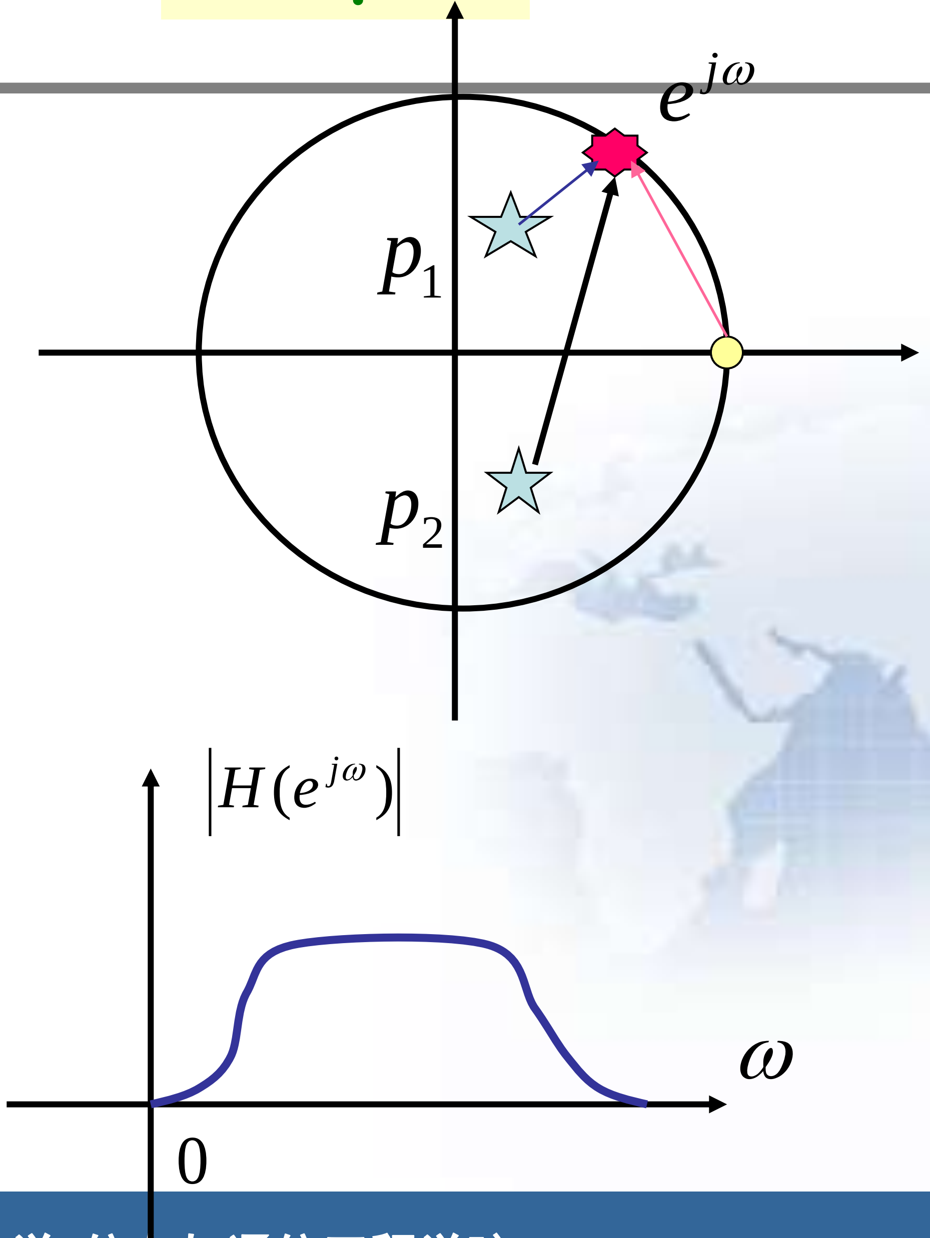
lowpass
$e^{j\omega}$
p_1
$\left|H(e^{j\omega})\right|$
0
ω

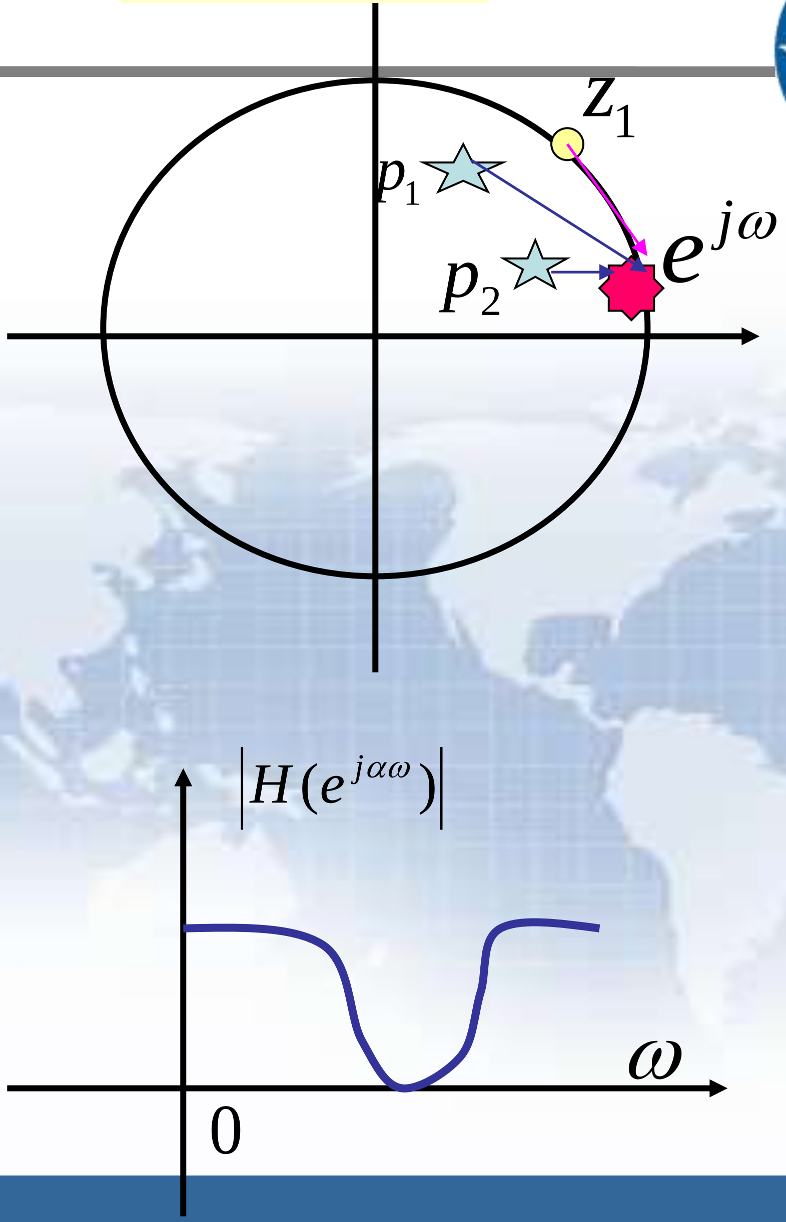


highpass
$e^{j\omega}$
p_1
z_1
$\left|H(e^{j\omega})\right|$
0
ω

bandpass

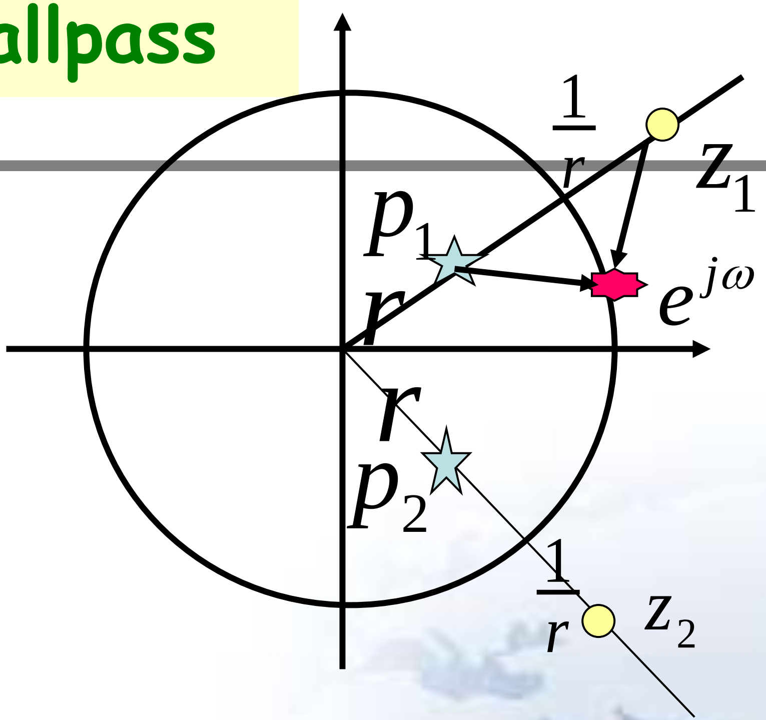
bandstop

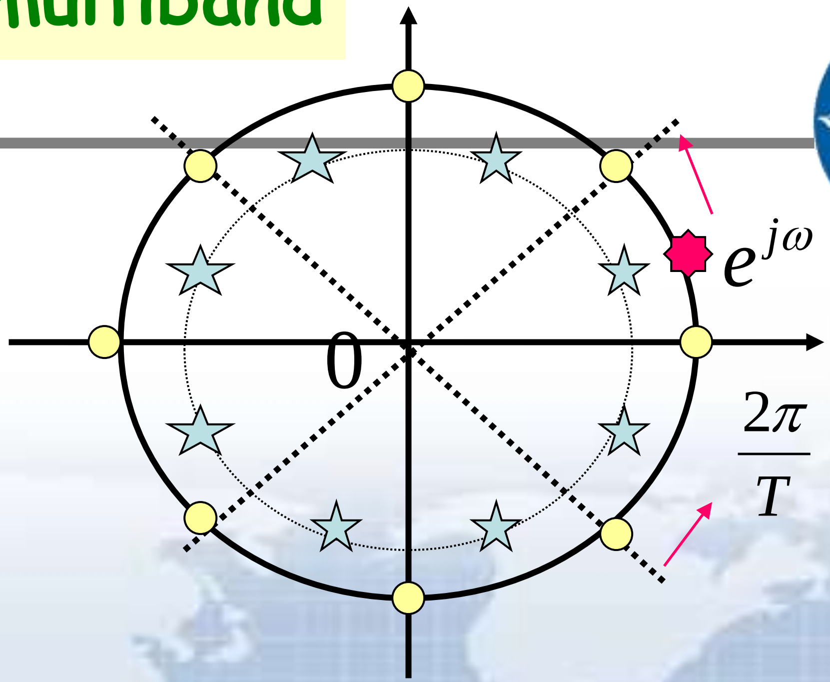


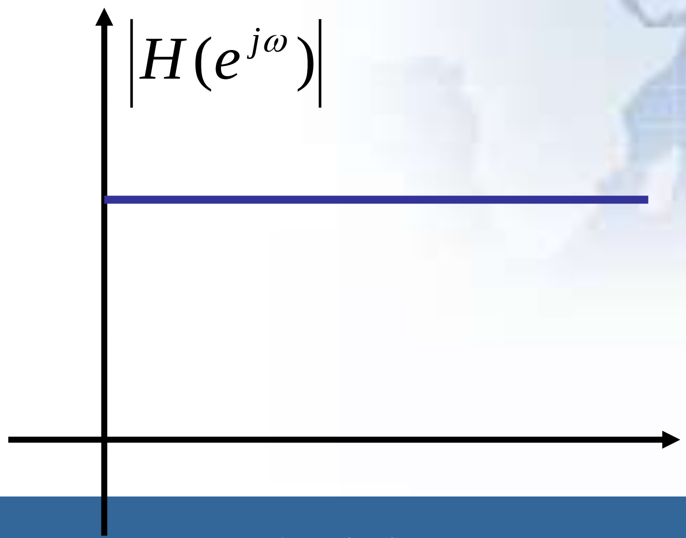


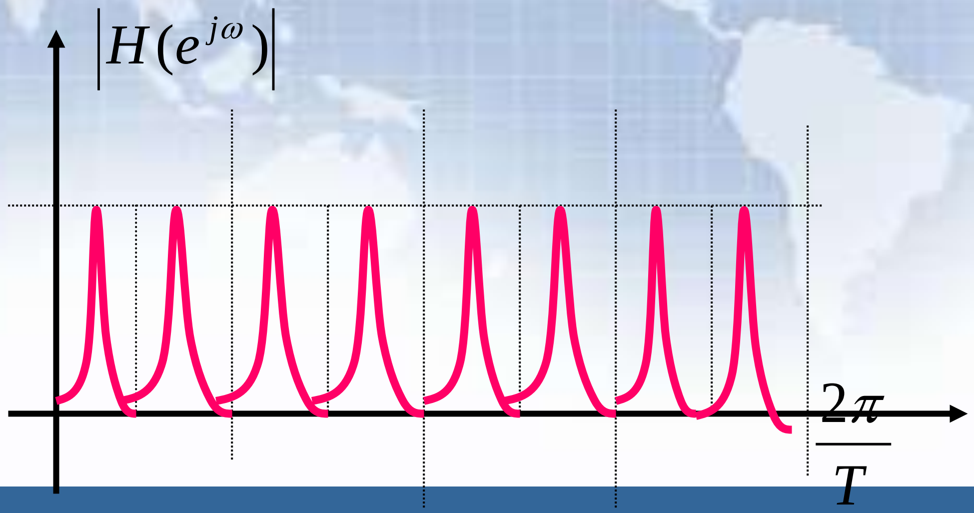
allpass

multiband









Given

$$\overrightarrow{c_m} = e^{j\omega} - c_m = \rho_m e^{j\theta_m}$$

$$\overrightarrow{d_k} = e^{j\omega} - d_k = l_k e^{j\phi_k}$$

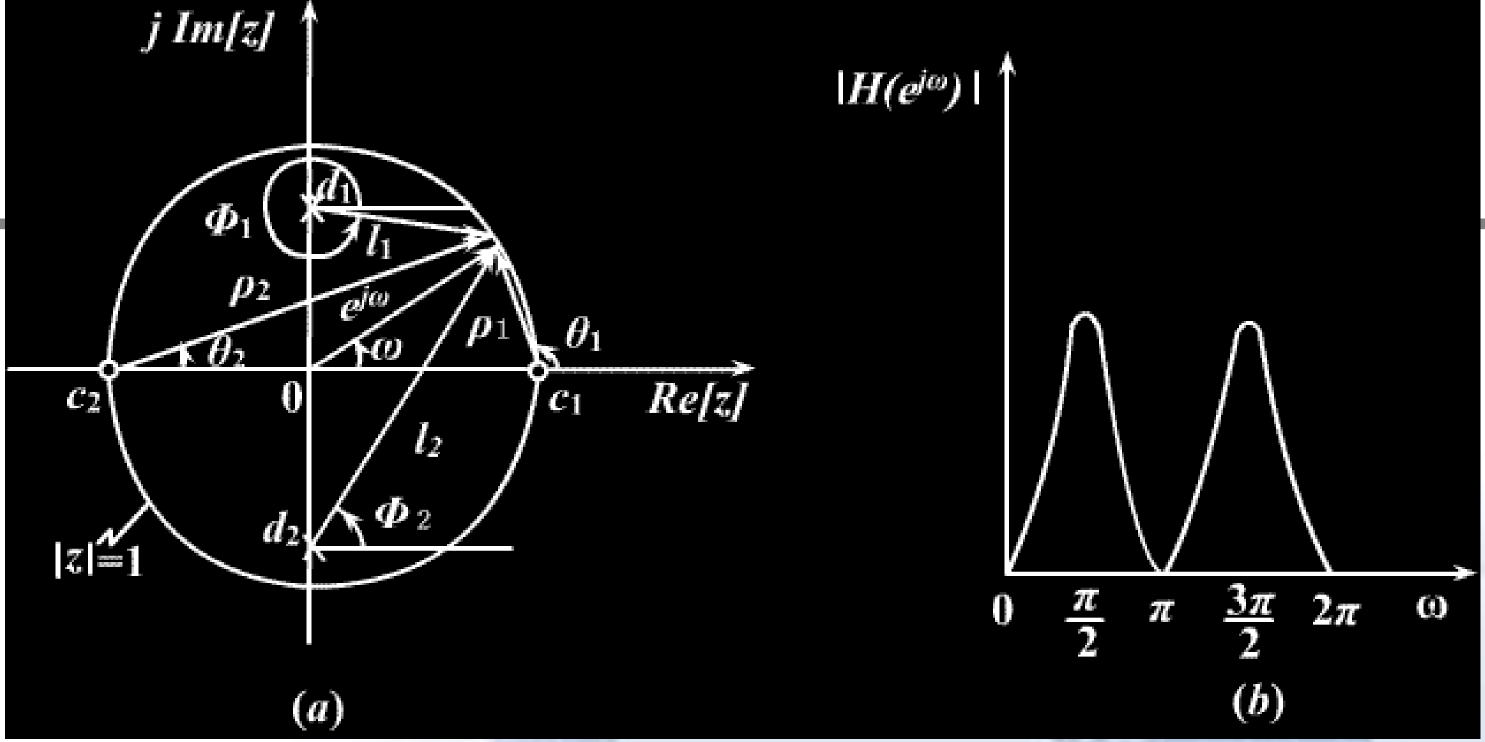
so, the frequency response

Amplitude :

$$\left|H(e^{j\omega})\right| = |K| \frac{\prod_{m=1}^{M} \rho_m}{\prod_{k=1}^{N} l_k}$$

Phase :

$$\arg[H(e^{j\omega})] = \arg[K] + \sum_{m=1}^{M} \theta_m - \sum_{k=1}^{N} \phi_k + (N-M)\omega$$



- Position of zeros will affect the position and depth of dips
  - Zeros at the unit cycle, the amplitude of dips will be zero
  - Zeros tend to the unit cycle, the amplitude of dips tend to zero
- Position of poles will affect the position and depth of peaks
  - Poles tend to the unit cycle, the amplitude of peaks tend to infinite
  - Poles is outside the unit cycle, the system is unstable

6.7.5 Stability Condition in Term of Pole Locations

- Before the digital filter is implemented, we need to ensure that the transfer function derived will lead to a stable structure.
- An LTI digital filter is BIBO stable if and only if:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

6.7.5 Stability Condition in Term of Pole Locations

So, the ROC of the transfer function $H(z)$ of a BIBO stable LTI digital filter should contains the unit circle.

The causal LTI digital filter is stable with all its poles are inside the unit cycle.

6.7.5 Stability Condition in Term of Pole Locations

- For a causal LTI FIR digital filter, the stability is always satisfied.
- For a causal LTI IIR digital filter, it may be unstable if not designed properly.
- In addition, due to the unavoidable quantization of all coefficients, an originally stable IIR filter may also become unstable.(P319 Eg6.41)

Homework

- Read textbook from p277 to 319
- Problems

6.6, 6.7,6.12,6.18,6.42

6.44,6.47,6.65,6.66,6.81

M6.1, M6.2, M6.3

THANKS